

Subject Description Form

Subject Code	EEE525
Subject Title	Smart Transportation for Green Cities
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. Understand the concepts of smart transportation and its role in the sustainability of urban mobility systems. 2. Learn about various technologies and solutions enabling smart and green city transportation. 3. Evaluate how intelligent traffic management, public transit optimisation and shared mobility can reduce emissions and congestion.
Subject Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Understand concepts of smart transportation technologies and strategies for sustainable urban mobility. b. Assess the role of data analytics and emerging technologies in optimising transportation systems for efficiency and environmental friendliness. c. Explore smart transportation systems' various components and technologies, such as intelligent traffic management and electric vehicles.
Subject Synopsis/ Indicative Syllabus	1. Introduction to Smart Transportation and Green Cities: Definition and scope of smart transportation, importance and benefits of smart transport systems, and the concepts and principles of green cities and sustainable urban development 2. Data Collection and Data Analytics: Introduction to the collection and analysis of transportation data, data sources, and big data analytics for optimising transportation systems. 3. Traveller Information Systems: Understanding the benefits of providing information to travellers, including estimating and predicting travel times, smart parking technologies - availability information, and real-time passenger information. 4. Traffic management with ITS: Applications of ITS in managing traffic on motorways and arterial roads, such as ramp metering, variable speed limits, lane usage management, electronic toll collection, public transport priority and incident detection. 5. Electric vehicle infrastructure: electric and hybrid vehicles, alternative fuels, EV charging infrastructure and scheduling. 6. Shared Mobility Services: car sharing, bike/scooter sharing, ride-hailing/sharing apps, on-demand rides, and MaaS for reducing personal vehicle ownership. 7. Connected Autonomous Vehicles and Cooperative ITS: The future of transportation with connected autonomous vehicles and the use of vehicle-to-vehicle, vehicle-to-infrastructure, and vehicle-to-everything (V2X) communication to enhance safety and efficiency.

	8. <i>Autonomous Metros Infrastructure</i> : driverless metro trains and the supporting control systems, and operation of an autonomous mass transit system designed for GoA 4 (Grade of Automation Level 4) standards.				
Teaching/Learning Methodology	Delivery of the subject is mainly through formal lectures and complemented by tutorials. The assignment provides students with hands-on experience in processing and analysing big data, while report-writing enables students to practise writing skills.				
	Teaching/Learning Methodology		Outcomes		
			a	b	c
	Lectures		✓	✓	✓
	Tutorials		✓	✓	✓
	Assignment				✓
Assessment Methods in Alignment with Intended Learning Outcomes					
	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed		
			a	b	c
	1. Mini Project	40%	✓	✓	✓
	2. Continuous Assessment	30%	✓	✓	
	3. Assignment	30%	✓	✓	✓
	Total	100%			
The mini project allows assessment of outcomes covering principles, techniques, design and application, supplemented by continuous assessments. Assignments enable students to explore and apply data analytics and evaluate transport system performance.					
Student Study Effort Expected	Class contact:				
	▪ Lecture/Tutorial			39 Hrs.	
	Other student study effort:				
	▪ Assignment			30 Hrs.	
	▪ Self-study			38 Hrs.	
	Total student study effort			107 Hrs.	
Reading List and References	Reference books: 1. US DoT, ITS ePrimer, ITS Joint Program Office, www.pcb.its.dot.gov/eprimer/ 2. PIARC, Cooperative Vehicle Highway Systems, Technical Committee 2.1 Road Network Operations, 2016. 3. R. Gordon, Intelligent Transportation Systems: Functional Design for Effective Traffic Management, Springer, 2016.				